Chapter 3
Vector Spaces

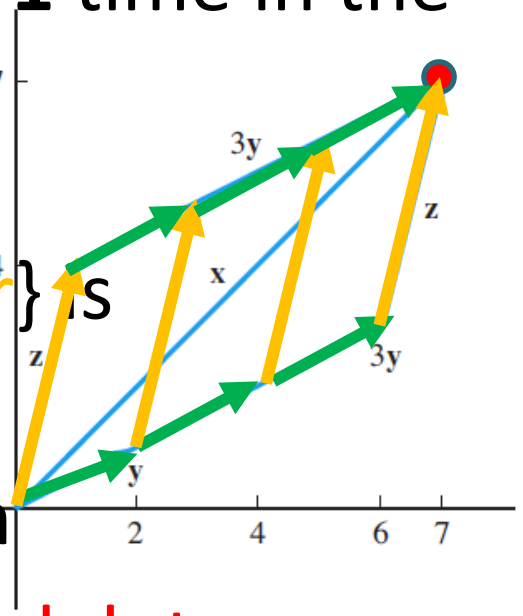
3.5 Change of Basis

Objective

- Concepts:
  - Given an ordered basis $E = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$, a vector $\boldsymbol{x}$ can be expressed as a **coordinate vector** $[\boldsymbol{x}]_E$.
  - A coordinate vector $[\boldsymbol{x}]_E$ in one basis $E = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$ can be written as another coordinate vector $[\boldsymbol{x}]_F$ in another basis $F = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots\}$.
- Calculations:
  - The **transition matrix** U from $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$ to the standard basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \dots\}$ is $U = [\boldsymbol{u}_1, \boldsymbol{u}_2, \dots]$.
  - The transition matrix from $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \dots\}$ to $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$ is U^{-1}.
  - The transition matrix from $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots\}$ to $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$ is $S = U^{-1}V$.

Motivation: describing a point

- Describe the red dot using 2 numbers.
- One way: x-coordinate: **7**. y-coordinate: **7**.
- Another way: the red dot is **3** times in the green vector direction, and **1** time in the yellow vector direction.



- {green vector, yellow vector} is a **basis** of $\mathbb{R}^2$
- $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ is a **coordinate vector** in that basis, describing the red dot.

Coordinate Vector

Definition

Let V be a vector space and let $E = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an ordered basis for V. If $\mathbf{v}$ is any element of V, then $\mathbf{v}$ can be written uniquely in the form

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$$

where $c_1, c_2 \dots, c_n$ are scalers. Thus, we can associate each vector $\mathbf{v}$ a unique column vector $\{c_1, c_2 \dots, c_n\}^T$. The vector is called the **coordinate vector** of $\mathbf{v}$ with respect to (w.r.t.) the ordered basis E, denoted $[\mathbf{v}]_E$.

Example: Coordinate in $\mathbb{R}^2$

- $E = \{\boldsymbol{e}_1 = \begin{bmatrix}1\\0\end{bmatrix}, \boldsymbol{e}_2 = \begin{bmatrix}0\\1\end{bmatrix}\}$ and $F = \{\mathbf{y} = \begin{bmatrix}2\\1\end{bmatrix}, \mathbf{z} = \begin{bmatrix}1\\4\end{bmatrix}\}$ are both basis of $\mathbb{R}^2$. Let $\mathbf{x} = \begin{bmatrix}7\\7\end{bmatrix}$.
- As a linear combination of $\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$: $\mathbf{x} = \begin{bmatrix}7\\7\end{bmatrix} = \mathbf{7}\begin{bmatrix}1\\0\end{bmatrix} + \mathbf{7}\begin{bmatrix}0\\1\end{bmatrix}$
  So, the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathrm{E} = \{\boldsymbol{e}_1 = \begin{bmatrix}1\\0\end{bmatrix}, \boldsymbol{e}_2 = \begin{bmatrix}0\\1\end{bmatrix}\}$ is $[\mathbf{x}]_E = \begin{bmatrix}7\\7\end{bmatrix}$
- As a linear combination of $\{\mathbf{y}, \mathbf{z}\}$: $\mathbf{x} = \begin{bmatrix}7\\7\end{bmatrix} = \mathbf{3}\begin{bmatrix}2\\1\end{bmatrix} + \mathbf{1}\begin{bmatrix}1\\4\end{bmatrix} = 3\mathbf{y} + \mathbf{z}$.
  So, the coordinate vector of $\mathbf{x}$ with respect to the basis $F = \{\mathbf{y} = \begin{bmatrix}2\\1\end{bmatrix}, \mathbf{z} = \begin{bmatrix}1\\4\end{bmatrix}\}$ is $[\mathbf{x}]_F = \begin{bmatrix}\mathbf{3}\\\mathbf{1}\end{bmatrix}$

Example

Let $V = \mathbb{R}^{2\times 2}, \mathbf{x} = \begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}$, and the ordered basis is

$$B = \left\{ \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \right\}.$$

Find $[\mathbf{x}]_B$.

Solution

- $\mathbf{x} = c_1 \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} =$
  $\begin{bmatrix} c_1 + c_3 + c_4 & -c_1 + c_2 \\ c_2 - c_4 & -c_3 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}$
- Solving the system of 4 equations: $c_1 = -1, c_2 = 2, c_3 = -2, c_4 = 4$
- Thus, the coordinate vector $[\mathbf{x}]_B = [-1,2,-2,4]^T$.

Changing Coordinates

- If we have a new basis, how can we find the new coordinate vector with respect to that basis?
- Given $\mathbf{u}_1 = \begin{bmatrix}3\\2\end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix}1\\1\end{bmatrix}$, consider the following two problems
  1. Given vector $\mathbf{x} = [x_1, x_2]^T$, find its coordinates w.r.t. $\mathbf{u}_1$ and $\mathbf{u}_2$.
  2. Given vector $c_1\mathbf{u}_1 + c_2\mathbf{u}_2$, find its coordinates w.r.t. $\mathbf{e}_1$ and $\mathbf{e}_2$.

Solution

- We will solve 2 first. $\mathbf{u}_1$=$3\mathbf{e}_1 + 2\mathbf{e}_2$, $\mathbf{u}_2$=$\mathbf{e}_1 + \mathbf{e}_2$, then

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 = (3c_1+c_2)\mathbf{e}_1 + (2c_1 + c_2)\mathbf{e}_2$$

- Thus, the coordinates of $c_1\mathbf{u}_1 + c_2\mathbf{u}_2$ w.r.t. $\{e_1, e_2\}$ is:

$$\mathrm{x}= \begin{bmatrix}3c_1 + c_2\\2c_1 + c_2\end{bmatrix} = \begin{bmatrix}3 & 1\\2 & 1\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix}$$

- Let $U = [\mathbf{u}_1, \mathbf{u}_2] = \begin{bmatrix}3 & 1\\2 & 1\end{bmatrix}$. To find the coordinate of $\mathbf{x}$, we simply multiply U with c: $\boxed{\mathbf{x} = U\mathbf{c}}$. U is called the *transition matrix*.

Changing Coordinates

- Solution to 1: As we have

$$\mathbf{x} = U\mathbf{c}$$

- Matrix U is nonsingular, therefore

$$\mathbf{c} = U^{-1}\mathbf{x}$$

Thus, U^{-1} is the transition matrix from $\{\mathbf{e}_1, \mathbf{e}_2\}$ to $\{\mathbf{u}_1, \mathbf{u}_2\}$.

Changing coordinates:
If $\mathbf{c}$ is the coordinate vector of $\mathbf{x}$ with respect to the basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots\}$, then

$$\mathbf{x} = U\mathbf{c}$$

And

$$\mathbf{c} = U^{-1}\mathbf{x}$$

where $U = [\mathbf{u}_1, \mathbf{u}_2, \dots]$ is called the **transition matrix** from the $\{\mathbf{u}_1, \mathbf{u}_2, \dots\}$ to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots\}$, and U^{-1} is the transition matrix from $\{\mathbf{e}_1, \mathbf{e}_2, \dots\}$ to $\{\mathbf{u}_1, \mathbf{u}_2, \dots\}$.

Example

Let $\mathbf{u}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$. Find the coordinate vector of $\mathbf{x}$ with respect to $\mathbf{u}_1$ and $\mathbf{u}_2$.

Example

Let $\mathbf{u}_1 = \begin{bmatrix}3\\2\end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix}1\\1\end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix}7\\4\end{bmatrix}$. Find the coordinate vector of $\mathbf{x}$ with respect to $\mathbf{u}_1$ and $\mathbf{u}_2$.

Solution

- The transition matrix from $\{\mathbf{u}_1, \mathbf{u}_2\}$ to $\{\mathbf{e}_1, \mathbf{e}_2\}$ is:

$$U = \begin{bmatrix}3 & 1\\2 & 1\end{bmatrix}$$

- The transition matrix from $\{\mathbf{e}_1, \mathbf{e}_2\}$ to $\{\mathbf{u}_1, \mathbf{u}_2\}$ is its inverse:

$$U^{-1} = \frac{1}{3-2}\begin{bmatrix}1 & -1\\-2 & 3\end{bmatrix} = \begin{bmatrix}1 & -1\\-2 & 3\end{bmatrix}$$

- Thus, the coordinates of vector $\mathbf{x}$ w.r.t. $\{\mathbf{u}_1, \mathbf{u}_2\}$ is

$$U^{-1}\mathbf{x} = \begin{bmatrix}1 & -1\\-2 & 3\end{bmatrix}\begin{bmatrix}7\\4\end{bmatrix} = \begin{bmatrix}3\\-2\end{bmatrix}$$

- (Meaning: $\mathbf{x} = 3\,\mathbf{u}_1 - 2\mathbf{u}_2$)

Changing Coordinates

Changing coordinate vectors from one basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ of $\mathbb{R}^2$ to another basis $\{\mathbf{u}_1, \mathbf{u}_2\}$:

- Suppose for a given vector $\mathbf{x}$, its coordinates w.r.t. $\{\mathbf{v}_1, \mathbf{v}_2\}$ are known:

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$$

- We want to write $\mathbf{x}$ as $d_1\mathbf{u}_1 + d_2\mathbf{u}_2$.

Then:

- We have $\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = d_1\mathbf{u}_1 + d_2\mathbf{u}_2$
- Let $V = [\mathbf{v}_1, \mathbf{v}_2]$ and $U = [\mathbf{u}_1, \mathbf{u}_2]$. Then,

$$V\mathbf{c} = U\mathbf{d}$$

$$\Rightarrow \mathbf{d} = U^{-1}\mathrm{V}\mathbf{c}$$

- $S = U^{-1}V$ *is the transition matrix* from $\{\mathbf{v}_1, \mathbf{v}_2\}$ to $\{\mathbf{u}_1, \mathbf{u}_2\}$.

Transition matrix between 2 basis

Changing coordinates:

If $\mathbf{c}$ is the coordinate vector of $\mathbf{x}$ with respect to the basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots\}$, and $\mathbf{d}$ is the coordinate vector of $\mathbf{x}$ with respect to the basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots\}$, then

$$\mathbf{d} = S\mathbf{c}$$

where $S = U^{-1}V$ is the **transition matrix** from $\{\mathbf{v}_1, \mathbf{v}_2, \dots\}$ to $\{\mathbf{u}_1, \mathbf{u}_2, \dots\}$

(new coordinate vector) = (transition matrix) times (old coordinate vector), so you need to remember how to calculate transition matrices.

Example (Textbook Example 4)

Find the transition matrix corresponding to the change of basis from $\{\mathbf{v}_1, \mathbf{v}_2\}$ to $\{\mathbf{u}_1, \mathbf{u}_2\}$, where $\mathbf{v}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$ and $\mathbf{u}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Example (Textbook Example 4)

Find the transition matrix corresponding to the change of basis from $\{\mathbf{v}_1, \mathbf{v}_2\}$ to $\{\mathbf{u}_1, \mathbf{u}_2\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 7 \\ 3 \end{bmatrix} \text{ and } \mathbf{u}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solution

- $U = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, $U^{-1} = \frac{1}{3\times 1 - 1\times 2}\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$
- $V = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix}$
- The transition matrix from $\{\mathbf{v}_1, \mathbf{v}_2\}$ to $\{\mathbf{u}_1, \mathbf{u}_2\}$ is

$$S = U^{-1}V = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -4 & -5 \end{bmatrix}$$

Property

- A transition matrix is nonsingular
- If S is the transition matrix from $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ to $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$, then S^{-1} is the transition matrix from $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ to $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

Example (textbook example 6)

Suppose that in P_3 we want to change the ordered basis $[1, x, x^2]$ to the ordered basis $[1, 2x, 4x^2 - 2]$.

Solution

Since $[1, x, x^2]$ is the standard basis for P_3, it is easier to find the transition matrix from $[1, 2x, 4x^2 - 2]$ to $[1, x, x^2]$.

$$1 = 1 \cdot 1 + 0x + 0x^2$$

$$2x = 0 \cdot 1 + 2x + 0x^2$$

$$4x^2 - 2 = -2 \cdot 1 + 0x + 4x^2$$

So the transition matrix is

$$S = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Example (textbook example 6)

The inverse of S will be the transition matrix from $[1, x, x^2]$ to $[1, 2x, 4x^2 - 2]$:

$$S^{-1} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}$$

(To calculate the inverse matrix, row reduce [S | I])

Example (textbook example 6)

Given any $p(x) = a + bx + cx^2$ in P_3, to find the coordinates of $p(x)$ with respect to $[1, 2x, 4x^2 - 2]$, we simply multiply

$$\begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a + \frac{1}{2}c \\ \frac{1}{2}b \\ \frac{1}{4}c \end{bmatrix}$$

Thus,

$$p(x) = \left(a + \frac{1}{2}c\right) \cdot 1 + \left(\frac{1}{2}b\right) \cdot 2x + \frac{1}{4}c \cdot \left(4x^2 - 2\right)$$

(Indeed, this is equal to $a + bx + cx^2$.)

Example

Let $B = \{x^2, x-1, 1\}$ and $C = \{x^2+x+1, x+1, 1\}$ be two basis for P_3. If $p(x) \in P_3$ with $[p(x)]_B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, find $[p(x)]_C$.

Solution

- (The standard basis is $\{1, x, x^2\}$) The transition matrix from B to C is

$$S = \left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 & -1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

- $[p(x)]_C = S[p(x)]_B = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$.

Summary

- Concepts:
  - **Coordinate vector** $[\boldsymbol{x}]_E$ tells you how to write a vector $\boldsymbol{x}$ as a linear combination of $\boldsymbol{u}_1, \boldsymbol{u}_2$,… in a basis E.
  - A coordinate vector $[\boldsymbol{x}]_E$ in one basis E can be written as another coordinate vector $[\boldsymbol{x}]_F$ in another basis F: new coordinate vector = **transition matrix** times old coordinate vector.
- Calculating transition matrices:
  - The transition matrix U from $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$ to the standard basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \dots\}$ is $U = [\boldsymbol{u}_1, \boldsymbol{u}_2, \dots]$.
  - The transition matrix from $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \dots\}$ to $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$ is U^{-1}.
  - The transition matrix from $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots\}$ to $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \dots\}$ is $S = U^{-1}V$.